CHAPTER 11

Science, Technology and Environment

The department deals with the subjects of popularisation of science, promotion of Science and Technology (S&T) projects relevant to the State, inculcation of environmental awareness and 'prevention/control' of environmental pollution and degradation. It acts as a nodal agency between the Government of India and the State Government in matters relating to Science, Technology and Environment. It also implements all measures connected with S&T applications through the respective line departments. The S&T as well as Environment programmes are undertaken independently and jointly through Government departments, Educational Institutions and NGOs. Preference is given to S&T projects in the field of municipal waste disposal/ treatment, recycling of waste agriculture, watershed management, aquaculture, bio diversity sustainable management of natural resources, etc.

The Integrated Mission for Sustainable Development (IMSD) project/data base is being utilized by various line departments, such as, Irrigation, Rural Development, Industries, Mines, P.W.D. etc. The state-of-the-art Geographical Information System (GIS) and Digital Image Processing (DIP) hardware and software, acquired through the grants obtained from the Department of Space (DoS), Government of India, are being used for remote sensing application projects as well as for National (Natural) Resources Information System (NRIS) database. Space Museum exhibits of INSAT/IRS Models acquired from the Department of Space (DoS), GoI, are being utilized for undertaking educational programmes for students and general public at large. A knowledge based approach scheme for remote sensing data utility is envisaged to identified line departments and academic institutions.

The National Natural Resources Management System (NNRMS) under the Department of Space have sanctioned Remote Sensing Training Programmes with different durations of 3 weeks, 5 days and 2 days for the University faculty and School/College Teachers from the Western Region, at an estimated cost of Rs.6.00 lakh. Grant-in-Aid is released to the Association of Friends of Astronomy (AFA) to maintain/run the astronomical observatories at Panaji, Ponda & Margao. Grants are also released to

deserving 'Institutions/NGOs' for undertaking science popularisation and applied S&T projects in the fields of entrepreneurial development, waste management, recycling, coastal zone management, invention of teaching aids and the like. Some common facilities required for R&D are installed in Government owned as well as other accredited Institutions. Training courses are conducted for 'scientific/technical' personnel involved in the implementation of schemes under the National Watershed Development Programme. Two mobile Scientific Education Vehicles (SEVs) which were launched in 2004-05 to give a boost to science education in remote/rural areas have proved to be most popular and are serving the education needs of rural as well as semi urban population of the State. A State-of-the-Art 3-D Theatre for Science related shows has been constructed at a cost of Rs. 67.50 lakh in the Goa Science Centre Complex at Miramar in collaboration with the National Council of Science Museums (NCSM).

Under the scheme Environmental Research and Ecological Regeneration, environmental awareness programmes and projects contributing to ecological 'conservation/ restoration' and adequate implementation of various environmental laws are undertaken. Environmental awareness programmes for the benefit of students and the community at large are undertaken in collaboration with the Directorate of Education and deserving NGOs. Projects conceived by educational/research Institutions and NGOs for identification and documentation of natural resources, environmental 'upgradation/ restoration', pollution control and ecological awareness programmes (including seminars & workshops) are promoted through this scheme. Annual programmes such as World Environment Day, Ozone Day and other events of environmental significance are observed. Support is given to research projects, material requirements and publicity needs contributing to adequate implementation of central and local environmental Acts/ Legislations. Similarly, publications of environmental relevance are sponsored.

Non-Conventional Sources of Energy

Under the scheme, various programmes involving new and renewable sources of energy have been implemented as under:

- Installation of Solar Water Heating Systems (SWHS) (500- Nos.) and Solar Cookers (150 nos.) under the domestic sector.
- Popularization and supply of Solar Photo-Voltaic (SPV) Systems. The targets achieved are; (i) solar lanterns 60 nos.; (ii) solar pumps 1 no.; solar street lights 70 nos.
- Popularization and installation of biogas plants under the centrally sponsored scheme. During the year, 100 biogas plants are in the process of being installed and commissioned at a subsidised rate.

Integrated Rural Energy Programme [IREP] is an area based programme with the block as a unit of planning to meet the energy needs for domestic and productive activities through the optimum use of both conventional and non-conventional energy sources and gadgets. Five blocks viz. Quepem, Sanguem, Sattari, Pernem and Canacona are covered under this programme. Subsidy is released for training, demonstration, publicity and supply of various non-conventional energy gadgets as well as energy saving devices in the blocks of Quepem and Pernem as per the recommendation of the energy survey already completed through an expert agency. The activities include supply of energy saving devices such as pressure cookers, release of connections for conventional electrification of remote hamlets and supply of renewable energy systems to supplement the energy requirements.

- The Wind Resource Assessment (WRA) study undertaken for the first time in the State of Goa in collaboration with the Center for Wind Energy Technology (CWET), Chennai, has been completed to the extent of 50% in South Goa district. The study in North Goa will start shortly after the installation of the wind mast in DSTE Complex, Saligao.
- The Detailed Project Report (DPR) for the proposed State Level Non-Conventional Energy park at Margao has been forwarded to the Ministry of Non-conventional Energy Sources, (GoI) for the release of grant of Rs.1.00 crore, as already committed for the said purpose by the Central Government. The proposal has been approved and the same is being set up near KTC Bus stand, Margao. The process of acquisition of land is completed.

Steps taken towards mobilization of resources

The Goa State Pollution Control Board (GSPCB), which is an autonomous statutory body under the administrative control of the department, earns sufficient revenue through consent fees to function with little or no financial support from the department. Similarly, it is creditable to note that the DSTE has been able to secure sufficient annual grants from the Central Government Ministries/Departments (Department of Science & Technology, Department of Space, Department of Ocean Development, Ministry of Non-Conventional Energy Sources and Ministry of Environment & Forests, Ministry of Rural Development Agency, to support various S&T projects as well as provide for the salaries of contractual staff working under the auspices of the Goa State Council for Science & Technology (GSCST). Funds have been mobilized through various Government of India schemes to complete major projects, such as the modern Sewage Treatment Plant at Panaji and the modern "GIS/DIP" facility at the DSTE.

Achievements

- The pioneering steps taken by the

department to control the proliferation of non-biodegradable plastic carry bags as well as to curb smoking and spitting offences have shown significant results. From April 2006 to September 2007, around 2660 cases have been booked and an amount of Rs. 2.70 lakh have been collected as fine under the Goa Prohibition of Smoking and Spitting Act. While, under the Goa Non Biodegradable Garbage (Control) Act, about 839 cases have been booked and an amount of Rs. 0.84 lakh have been collected as fine.

- Execution of the project entitled 'Critical Habitat Information System (CHIS)', sponsored by the ICMAM Directorate, Department of Ocean Development, (GoI), Chennai. The themetic maps of the project have been generated and are being integrated in a GIS platform.
- The following watershed development projects which were implemented through the GSCST and funded through Central grants are being continued during this year also
  :i) Morpirla Watershed Project (2,566 Ha) of five years duration with a sanctioned cost of Rs. 1.54 crore.
  ii) Tuyem Watershed Project (5,000 ha.) of five years duration with a sanctioned cost of Rs. 3.00 crore.
  iii) Kola Watershed Project (5,000 ha.) of five years duration with a sanctioned cost of Rs. 3.00 crore.

In addition, following two projects have been sanctioned recently:

i) Saligao Watershed Project (1340 ha.) of five years duration with a sanctioned cost of Rs.80.40 lakh.
ii) Morjim Watershed Project (1580 ha.) of five years duration with a sanctioned cost of Rs. 90.80 lakh.

- The Pilot 'Plasma Pyrolysis' Incinerator plant has been installed at Goa Medical College Hospital, Bambolim, in collaboration with DST (GoI), for safe 'incineration/disposal' of bio-medical waste generated in the Hospital, is working satisfactorily and is being monitored carefully for perfecting the state-of-the-art technology put to trial for the first time in the Country.
- Department has completed the National project entitled "National (Natural) Resources Information System (NRIS)" having a comprehensive digital database on natural resources of Goa (e.g. land, water, forests, environment, minerals, geology, soil, etc.) coupled with socio-economic data such as demography, general facilities (power/medical/educational), etc.
- Under the National Green Corps (NGC) programme, 552 Eco/Science/Nature clubs have been established through the support of the Ministry of Environment and Forests (MoEF), Government of India and also with the support of the State Government. This program has been conceived exclusively towards promotion of Environment awareness and its propagation. This scheme sensitizes and motivates the students

towards their concern for environment protection and conservation.

- A Patent Information Centre (PIC) established in the department with the support of the Department of Science & Technology (Government of India), has initiated its efforts to establish the Geographical Indication of the famous Goan beverage "Cashew Fenny". The application in this regard has been submitted to the Geographical Indication Registry, Chennai.
- The Goa State Biodiversity Board (GSBB) re-constituted in August 2007 under the Biological Diversity Act, 2003, has commenced its activities to start the groundwork towards the finalisation of needs of its office setup, drafting of Rules and preparation of five project proposals to study, review and document the biodiversity of Goa. The proposals have been submitted to the National Biodiversity Authority for funding support.
- The prestigious 12.5 MLD C-Tech, Sewage Treatment plant for the Capital City of Panaji has been completed at a cost of Rs. 14.10 crore, out of which Rs.9.87 crore were secured as grant from the Ministry of Environment & Forests (GoI) under the National River Conservation Project. This is the first Sewage Treatment Plant of its kind in the country and has been established in collaboration with the P.W.D.
- The department through (GSCST) has succeeded in securing the sanction of the Central Government for the execution of following new projects:

1. Prioritization of Wetlands in the State of Goa & their Potential Sustainable Evaluation using Remote Sensing Data.
2. Disaster Awareness, Preparedness, Mitigation and Management, Training Workshop in Goa.

- The department was instrumental in organizing the Children's Science Congress 2007, in November 2007, and Science Fiesta, 2007 in February 2007 in collaboration with the NCSTC.
- A project entitled "Mapping of S&T needs of the Goa State" is being executed with a prime objective to establish a central database repository on available scientific resources of the State so as to enable to adopt appropriate S&T interventions to mitigate S&T interventions.
- GSCST is engaged in setting up an ENVIS centre in collaboration with the MoEF by creating an interactive website on status of environment and other environment related issues through local language interface.
- GSCST, as the State Host Institute, (SHI) is in the process of publishing a State of Environment (SoE) report first of its kind on two focal themes namely 'Water Resources' and 'Coastal Land Resources' of Goa.
- GSCST has established a Edusat a talk back terminal facility in collaboration with the Department of Science and Technology (DST), GoI, New Delhi,

Indian Space Research Organisation (ISRO) and Vigyan Prasar. This has enabled the students/NGOs/General public to have on-line video conferencing on general science related issues/updates.

- Conducted Environmental Awareness Drives Clean Goa Green Goa.
- Domestic Waste Management and Vermicomposting training programme.
- District and State level Children Science Congress.
- Propogation of Science and Technology through EDUSAT.
- A land degradation project on 1:50,000 scale has been sanctioned to the GSCS&T by the National Remote Sensing Agency (NRSA) Department of Space, Government of India.

GOA STATE POLLUTION CONTROL BOARD

The Goa State Pollution Control Board established its first Regional Office in Goa in the quarters of the Industrial Development Corporation at Verna Industrial Estate, Verna Goa.

A new mobile Laboratory is made operational from January 2007 which is fitted with instruments to determine pollutants. It is extensively used for inspections and monitoring purposes.

A large electronic Display Board is set up at the entrance of the Kadamba Bus stand, Panaji in collaboration with Indian Environmental Association (IEA). The Board continuously displays the concentration levels of SO_2, NOx, SPM and RSPM in ambient air measured twice a week. Similar display boards are set up at M/s Sesa Goa, M/s Syngenta and M/s Ciba Speciality. It is now planned to set up such display boards at Zuari Nagar. This board will also display the concentration levels of ammonia in the ambient air.

The Board Laboratories are well equipped and fully functional to carry out the analysis of water, air and soil samples for physical and bacteriological parameters. A Graphite Furnace accessory for the existing Atomic Absorption Spectrophotometer (AAS) and a Gas Chromatograph were procured and installed in the Board laboratory in 2006. The instruments are being used for measurement of levels of toxic metals and pesticides in water/sediment samples.

The Board has set up an Automatic Weather Monitoring Station in Panaji which gives metereological data such as wind direction, wind speed, temperature, humidity, solar radiation and rainfall at an interval of every hour. The data generated can be used to understand the movements and dispersion of pollutants through modeling studies.

The Central Pollution Control Board has sanctioned an amount of Rs. 6.21 lakh for construction of Demo Road from Plastic waste of Ponda Municipal Council. The Ponda Municipality has agreed to lay down the Plastic road of about one kilometer in length.

The process for laying a road using plastic waste involves the following steps:

i) The plastic waste is segregated and shredded into small pieces
ii) The solid aggregate (granite, ceramic) with proper proportion of metals having different size and dust (quarry dust) is heated to around 170^0 C in Mini Hot Mix Plant. Simultaneously, Bitumen is heated at around 160^0 C and mixed.
iii) The heated aggregate is continuously stirred. Shredded plastic is added to get an uniform coating at the surface of the aggregate.
iv) Bitumen is added to aggregate coated with plastic waste and mixed uniformly as per IRC specifications. The mix is referred to as Waste Plastic-Aggregate-Bitumen Mix.

Sampling and analysis of industrial effluents

The Board has established its own laboratories which are well equipped with modern instrumentation to carry out water, air and soil analyses for physical, chemical and microbiological parameters. During the period April 2006 to March 2007, the Board has collected 374 samples of industrial effluents and analyzed the same for the following parameters: ph, Biochemical Oxygen Demand, Chemical Oxygen Demand, Suspended solids and Oil and Grease. Apart from these, the Board has collected and analyzed fresh water samples from different sources such as wells, springs, streams etc. A total of 108 sediment samples and 147 water samples have been analysed during the year for trace and heavy metals using the Atomic Absorption Spectrophotometer.

For the first time, the Board has started monitoring on its own the ambient air and source emissions from stacks for industries in Goa.18 ambient air and stack monitoring programmes were carried out during the year. Samples were collected and analysed for air pollutants such as SPM, RSPM, SO_2 and NOx for ambient air and particulate matter for stacks.

For collection and analysis of air, water and sediment samples, fees are being charged as per the guidelines of the Central Pollution Control Board. The total revenue generated during the year is Rs. 2.23 lakh towards analysis of samples.

National Water Quality Programme (NWMP) : The Board monitors the water quality of the rivers viz. Mandovi at Panaji, Mandovi at Tonca Marcela, Zuari at Cortalim, Zuari at Panchawadi, Khandepar at Opa, Ponda, Madei at Dabose,Valpoi, Valvanti at Sanquelim, Assanora at Assanora, Kalna at Chandel Pernem, Talpona at Canacona and Salaulim dam. 124 samples were collected from all these rivers and analyzed during the year.

National Air Monitoring Programme (NAMP): The Board monitors the ambient air quality at Vasco, MPT at Mormugao and

Panaji. The samples were collected at each station twice a week as per the standard norms and normal parameters like RSPM, SPM, SO_2 and NOx were analysed.

Spatial Environmental Planning Programme: The Spatial Environment Planning Programme was conceptualized for ensuring the protection of the environment and its resources through planned and sustainable development. The activities viz. Preparation of State Environmental atlas, revision of zoning atlas for siting of industries for North and South Goa, preparation of Industrial siting guidelines for North and South Goa districts.

Implementation of Hazardous Waste (Management & Handling) Rules, 1989 & amendment: At present, the Board has identified 198 hazardous waste generating industries and authorizations have been granted to these units.

Implementation of Bio Medical Waste (Management & Handling) Rules, 1988: As on March 2007, 171 Hospitals and 11 Veterinary units have been granted authorizations.

Noise Pollution Regulation & Control Rules, 2000: The Board is monitoring noise levels where specific complaints of noise pollution are received and action is being taken accordingly. The Government of Goa has designated District Magistrates, Director General of Police, Dy. Collector and Sub Divisional Magistrates, Superintendent of Police and Sub Divisional Police officers as authority for the maintenance of ambient air quality standards in respect of noise.

Implementation of Municipal Solid Wastes (Management & Handling) Rules, 2000: The Board has issued authorization to all the 14 local bodies under the Municipal Solid Waste (Management & Handling) Rules. Five Municipal bodies have partly adopted effective composting methods for processing of their waste.